

CS1CA Computer Systems Architecture

Instructions – machine language (1)

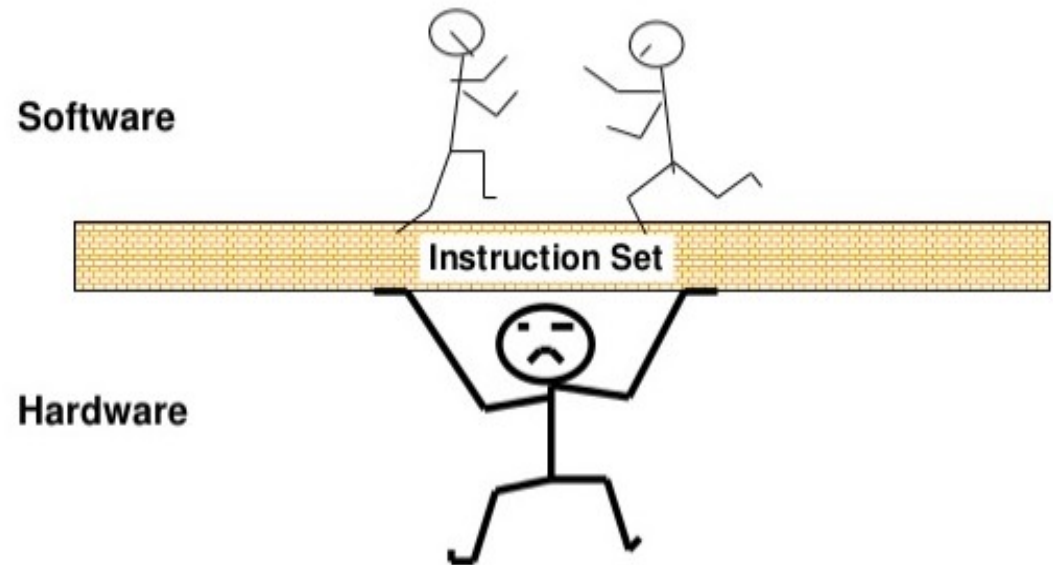
Week 5 lecture 1 of autumn 2025

Lectured by

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(Figure: courtesy of the Internet)

This topic will be delivered
in 4 lectures of weeks 5&7, and focus
on instruction set architecture

Outline of contents under this topic

- General introduction
- Classifying instruction set architecture
- Components in instructions
 - Memory addressing
 - Operands in instruction sets
 - Operations in instruction sets
- Encoding instruction sets
- A model machine: DLX architecture

In this lecture

- The following contents will be delivered
 - General introduction to instruction set architecture
 - Classifying instruction set architecture
- The learning outcomes are listed below
 - To be able to give the definition of instruction set architecture
 - To be able to articulate advantages and disadvantages of different classes of instruction set architecture

What is meant of instruction set architecture?

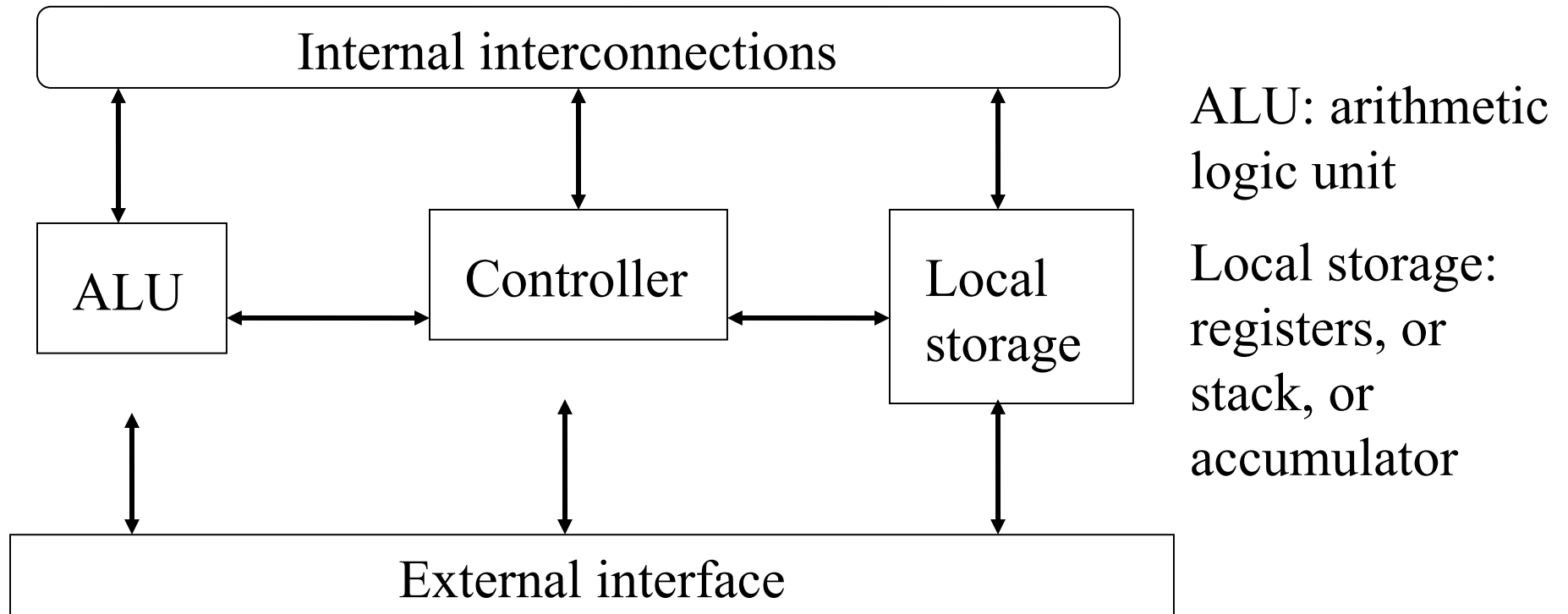
- Instruction set architecture is the portion of the machine visible to the programmer or compiler writer.
- Instructions are *binary combination* to represent
 - Registers
 - Addressing modes
 - Operands and operations

For example: in the 16-bit instruction “Add R1, R2, R3” used in the simple machine, 4 bits for opcode, 12 bits for operands, which are the three registers in this case.

0101 0001 0010 0011 => 0x5123
Add R1 R2 R3

Illustration of a processor

In order to design instruction sets, it is useful to understand the CPU organisation. Let's re-visit it.



The major conceptual units in a conventional processor

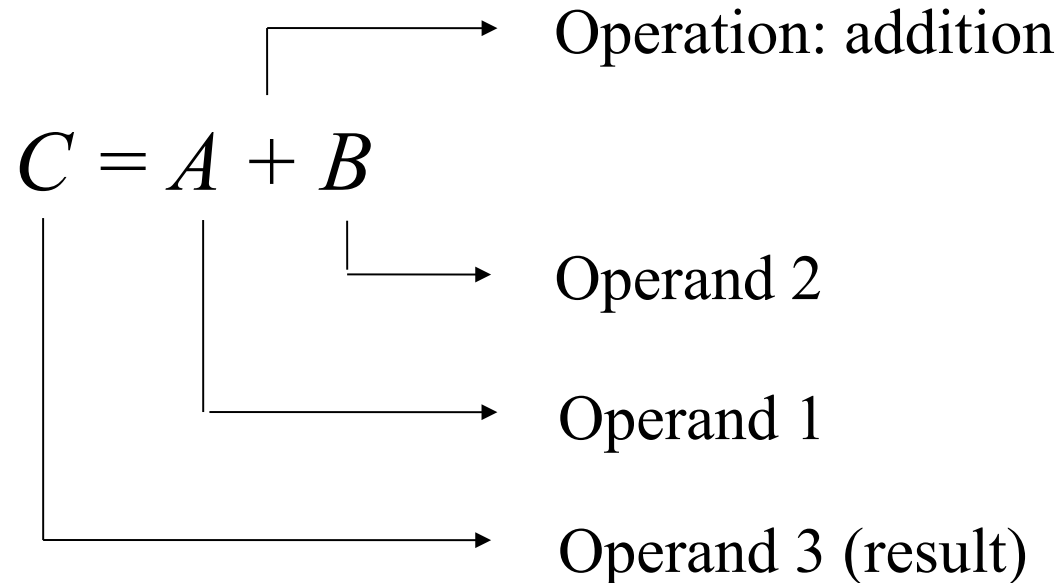
Quiz



Classification of instruction set architecture

- Stack architecture
- Accumulator architecture
- Register-memory architecture
- Register-register architecture (load-store)
- Memory-memory architecture (no long used)

Example: The task is to find out the solution of $A+B$.



Algorithm:

Step 1: load A

Step 2: load B

Step 3: add A and B

Step 4: store C

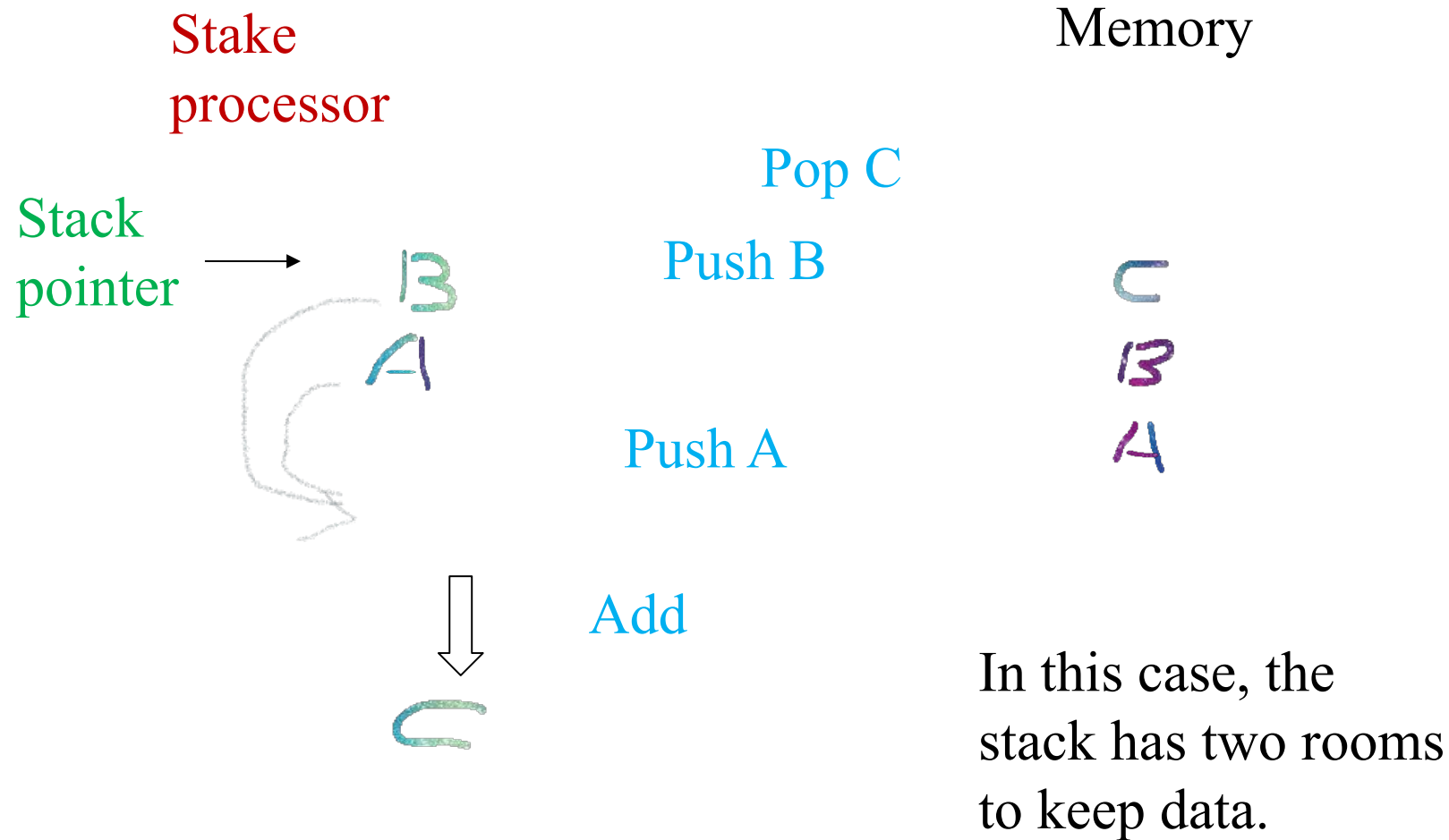
Using this simple example, let's see how it is implemented in the four different classes of instruction set architecture.

Example: The four instruction set architecture classes
The code sequence for purpose of $C = A + B$

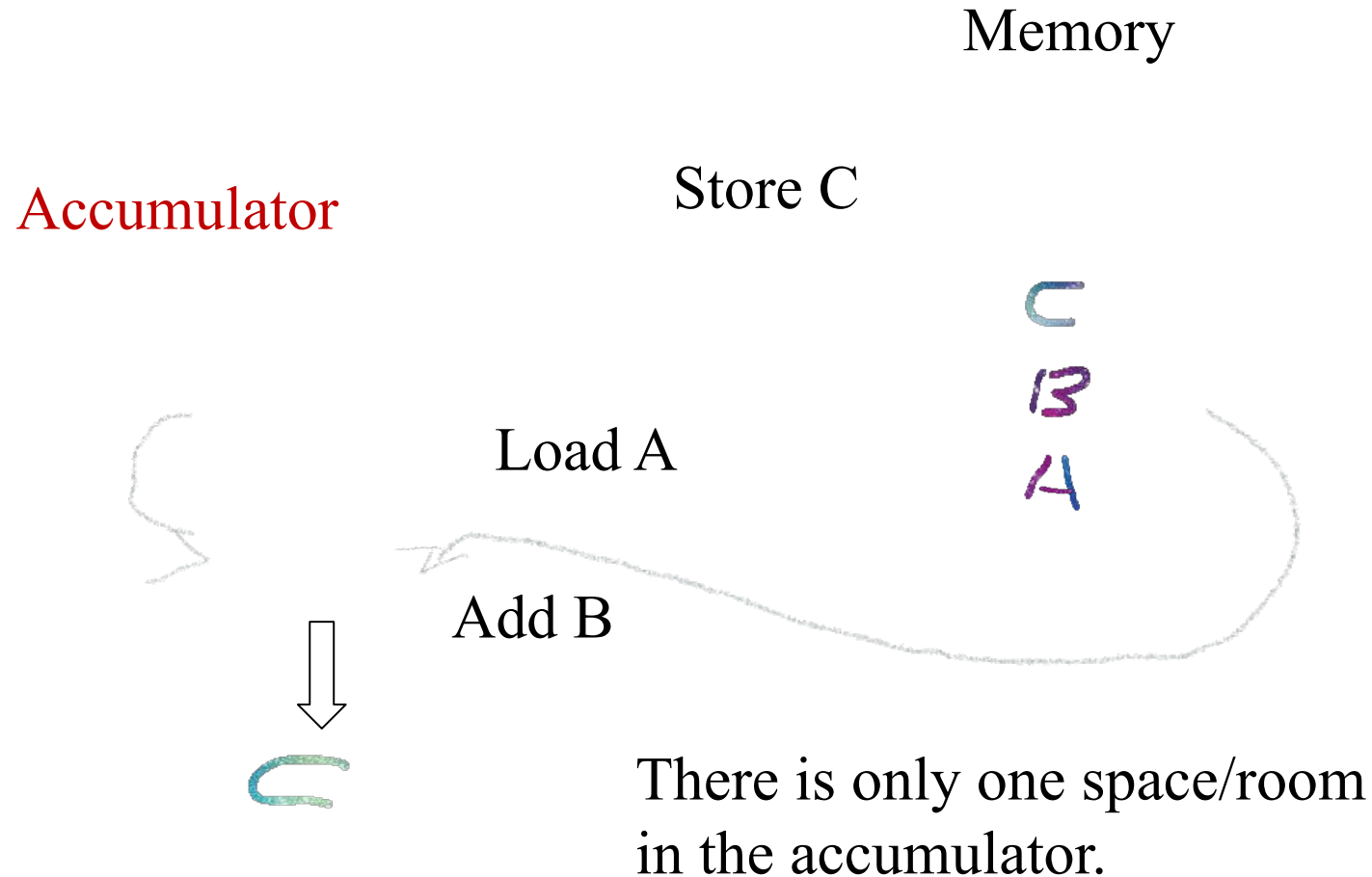
Stack	Accumulator	Register-memory	Register-register (load-store)
Push A	Load A	Load R1, A	Load R1, A
Push B	Add B	Add R1, B	Load R2, B
Add	Store C	Store C, R1	Add R3, R1, R2
Pop C			Store C, R3

- **Before 1980:** most machines used stack and accumulator architectures
- **After 1980 till now:** most machines use load-store architecture

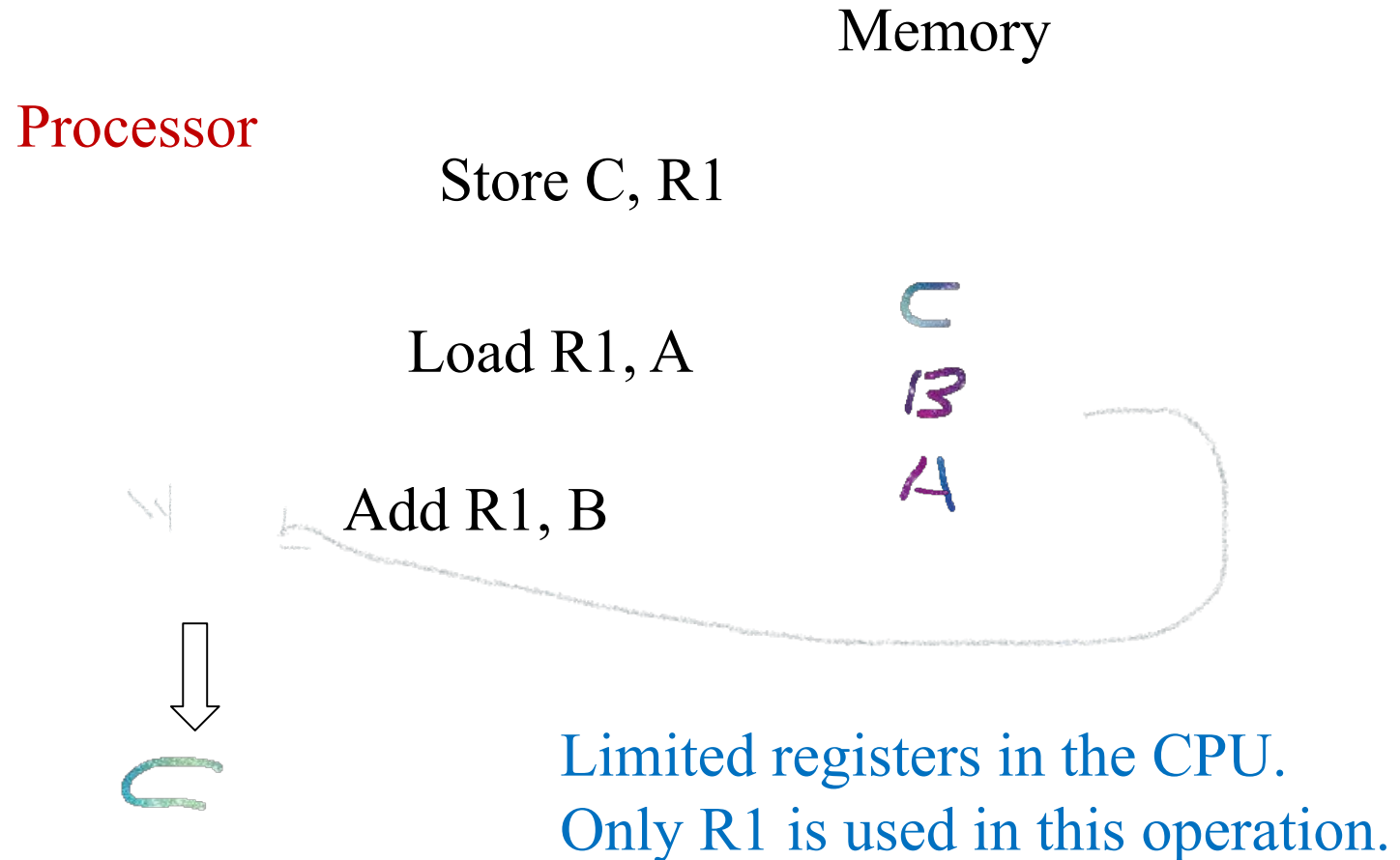
For the stack architecture



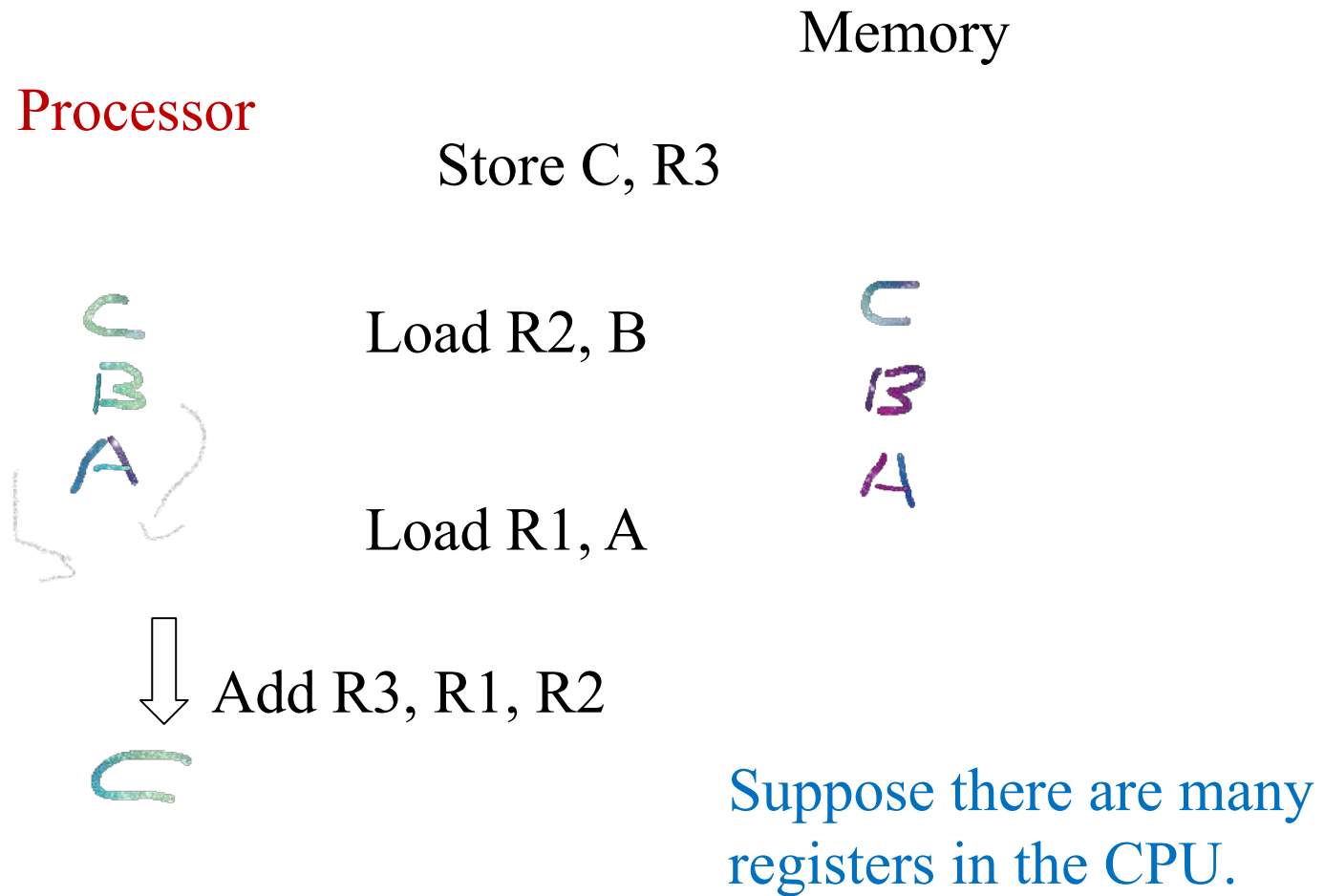
For the accumulator architecture



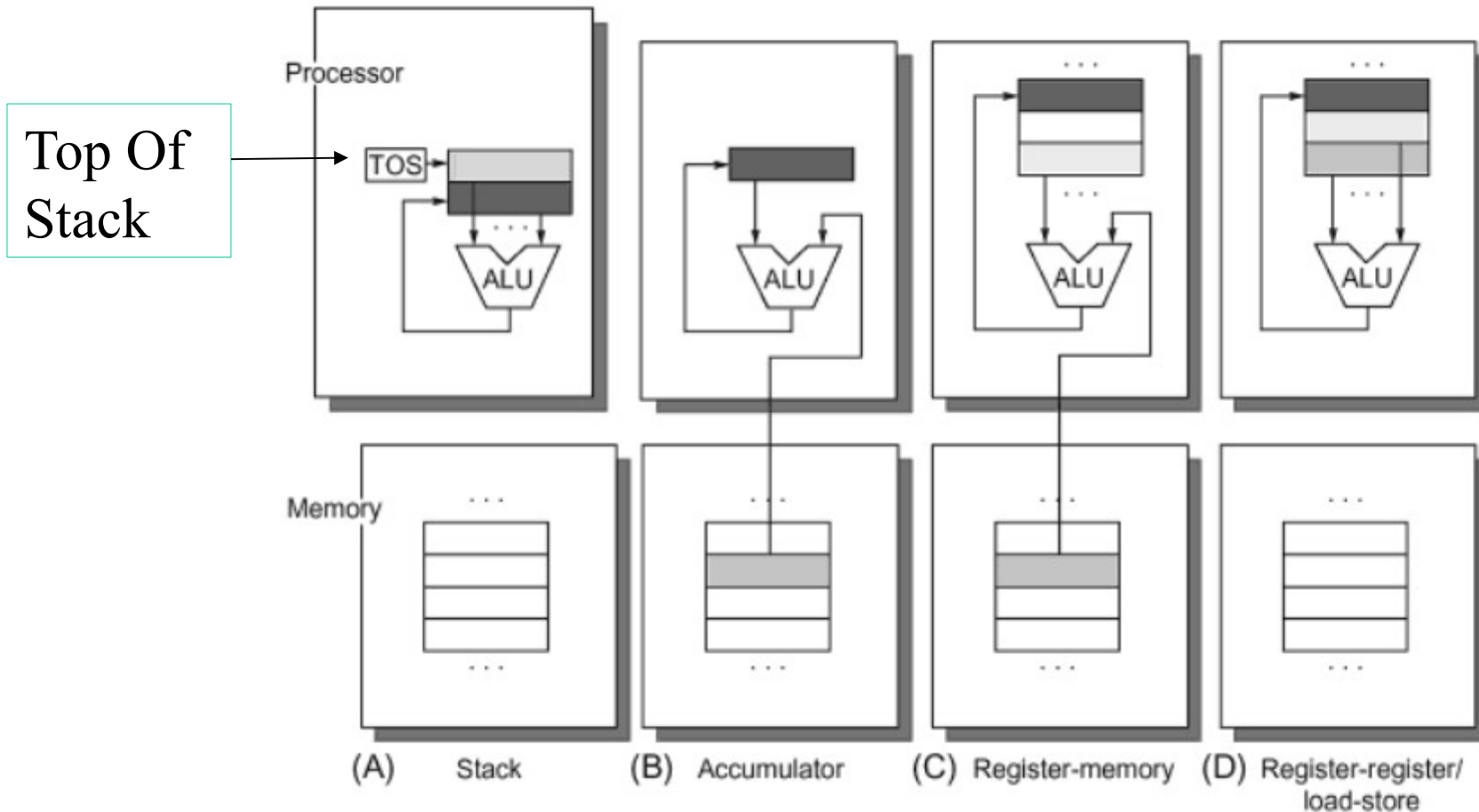
For the register-memory architecture



For the register-register (load-store) architecture



Operand locations for the four classes of instruction sets in the example



Advantages and disadvantages

	Register-register	Register-memory
Advantages	• Simple, fixed-length instruction encoding; • Simple code-generation model; • Instructions take similar number of clock cycles to execute.	• Data can be accessed without loading first; • Instruction format tends to be easy to encode and yield good density.
Disadvantages	• Higher instruction count; • Bit encoding can be wasteful because of fixed length encoding.	• Clock cycles per instruction varies; • Restrict number of registers for memory operations.

End of the lecture

Questions and exercises

- What is instruction set architecture?
 - Definition
 - Components involved in it
- Compare and contrast Register-register and Register-memory instruction set architectures
 - Using the example of $C = A + B$, explain what instructions are involved in the operations. Illustrate these operations with data manipulated in registers/stack/accumulator and memory cells
 - Identify advantages and disadvantages of the two different architectures, register-register and register-memory, in designing instruction sets